

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458101.6399 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.186 +/- 0.014
Transit depth (Rp/Rs)^2	3.46 +/- 0.52 [%]
Orbital Inclination (inc)	84.15 +/- 0.43
Airmass coefficient 1 (a1)	1.0086 +/- 0.0093
Airmass coefficient 2 (a2)	-0.0055 +/- 0.0036
Scatter in the residuals of the lightcurve fit is	0.92 %
Best Comparison Star	#3 - [225, 341]
Optimal Aperture	3.72
Optimal Annulus	5.99
Transit Duration (day)	0.0695 +/- 0.0037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.186 ± 0.014 vs 0.1646 (deviation 1.53σ)
Duration fit vs published	0.0695 d vs 0.0758 d (deviation 1.71σ)
Mid-time O–C	-4.6 min
Depth SNR	13.3
Residual scatter	0.92 %

Light curve

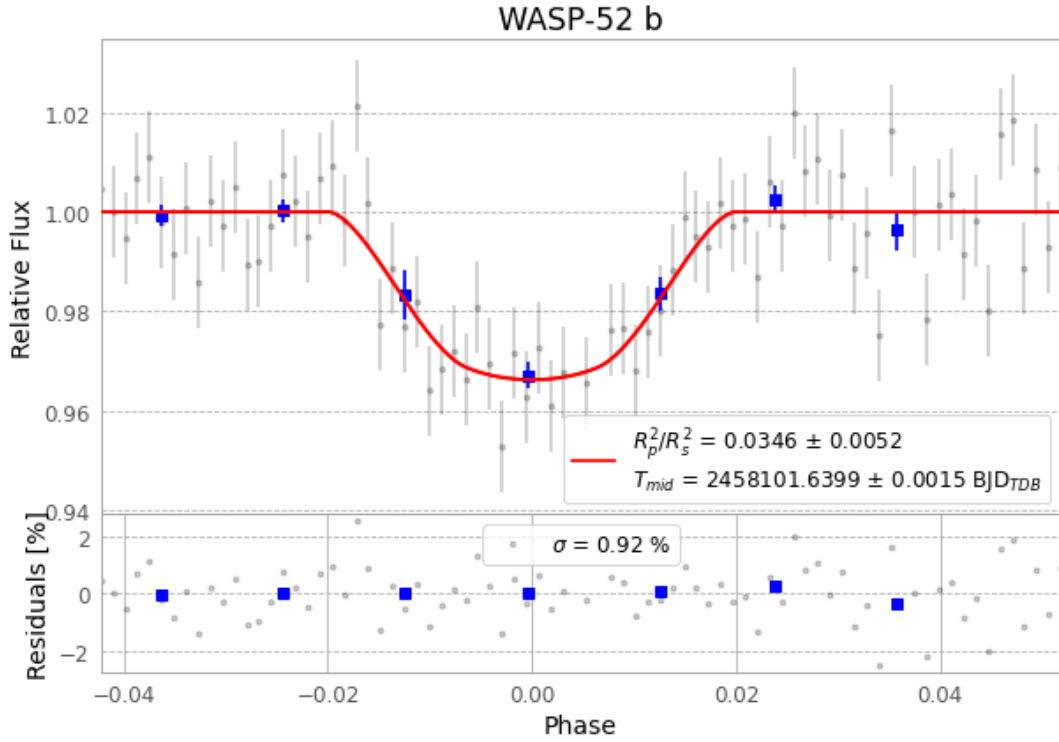


Figure 1 — FinalLightCurve_WASP-52 b_2017-12-13.png (SHA-256 558d15d4fbc163d4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [249, 269], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 aa1ffea7f1a13919...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

80 FITS frames (53 MB), WASP-52171214013312.FITS → WASP-52171214053012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-171214.log (748366925ddd...), FinalLightCurve_WASP-52 b_2017-12-13.png (558d15d4fbc1...), FinalLightCurve_WASP-52 b_2017-12-13.csv (03c2894288fa...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 09:52Z.